Hyperspectral Remote Sensing Images Salient Object Detection: The First Benchmark Dataset and Baseline

Peifu Liu, Huiyan Bai, Tingfa Xu, Jihui Wang, Huan Chen, and Jianan Li

***Abstract*— The objective of hyperspectral remote sensing image salient object detection (HRSI-SOD) is to identify objects or regions that exhibit distinct spectrum contrasts with the background. This area holds significant promise for practical applications; however, progress has been limited by a notable scarcity of dedicated datasets and methodologies. To bridge this gap and stimulate further research, we introduce the first HRSI-SOD dataset, termed hyperspectral remote sensing saliency dataset (HRSSD), which includes 704 hyperspectral images (HSIs) and 5327 pixel-level annotated salient objects. The HRSSD dataset poses substantial challenges for SOD algorithms due to large-scale variation, diverse foreground–background relations, and multisalient objects. Additionally, we propose an innovative and efficient baseline model for HRSI-SOD, termed the deep spectral saliency network (DSSN). The core of DSSN is the cross-level saliency assessment block (CSAB), which performs pixel-wise attention and evaluates the contributions of multiscale similarity maps at each spatial location, effectively reducing erroneous responses in cluttered regions and emphasizes salient regions across scales. Additionally, the high-resolution fusion module (HRFM) combines bottom–up fusion strategy and learned spatial upsampling to leverage the strengths of multiscale saliency maps, ensuring accurate localization of small objects. Experiments on the HRSSD dataset robustly validate the superiority of DSSN, underscoring the critical need for specialized datasets and methodologies in this domain. Further evaluations on the HSOD-BIT and HS-SOD datasets demonstrate the generalizability of the proposed method. The dataset and source code are publicly available at https://github.com/laprf/HRSSD**

***Index Terms*— Hyperspectral remote sensing images, hyperspectral remote sensing saliency dataset (HRSSD) dataset, salient object detection (SOD), spectral saliency.**

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I. Introduction

HYPERSPECTRAL salient object detection (HSOD) aims to identify objects or regions within a scene that exhibit distinct spectral characteristics compared to their surroundings [1], [2], [3]. Hyperspectral images (HSIs) capture detailed spectral reflectance properties, enabling differentiation of materials based on their unique spectral signatures [4], [5]. This capability provides HSIs with a significant advantage over RGB images, particularly in scenarios where foreground (orange star) and background (blue star) objects share similar visual appearances but differ spectrally as shown in Fig. 1.

Remote sensing images, acquired from satellites, drones, or other spacecraft, offer a bird's-eye view of Earth's surface. Despite the various application scenarios for hyperspectral remote sensing image salient object detection (HRSI-SOD), such as military defense [6], mineralogical mapping [7], and atmospheric monitoring [8], research in this domain has predominantly focused on natural scenes [1], [2], [3], [9], [10]. This trend can be attributed to two main factors: 1) the scarcity of publicly available datasets tailored for HRSI-SOD and 2) the lack of specialized methodologies addressing the unique challenges of this domain.

To advance this field, we introduce the hyperspectral remote sensing saliency dataset (HRSSD), the first benchmark dataset specifically designed for HRSI-SOD. Derived from WHU-OHS [11], HRSSD comprises 704 HSIs, each with a spatial resolution of 512 × 512 pixels and 32 spectral channels spanning the 466–940 nm wavelength range, accompanied by pixel-level annotations for 5327 salient objects. This dataset's broad coverage and varied land cover types present three primary challenges for SOD.

1) *Large-scale variation:* Spatial scales of land cover differ substantially, leading to significant variation in object sizes across images. Additionally, the spatial distribution of the same land cover type varies widely across regions, increasing the diversity of salient object sizes within a single image.
2) *Diverse foreground–background relations:* The spatial distribution and spectral characteristics of land cover, combined with its differing roles across various environments, render certain land covers salient in specific scenes, while they appear as background in others, complicating foreground–background delineation.

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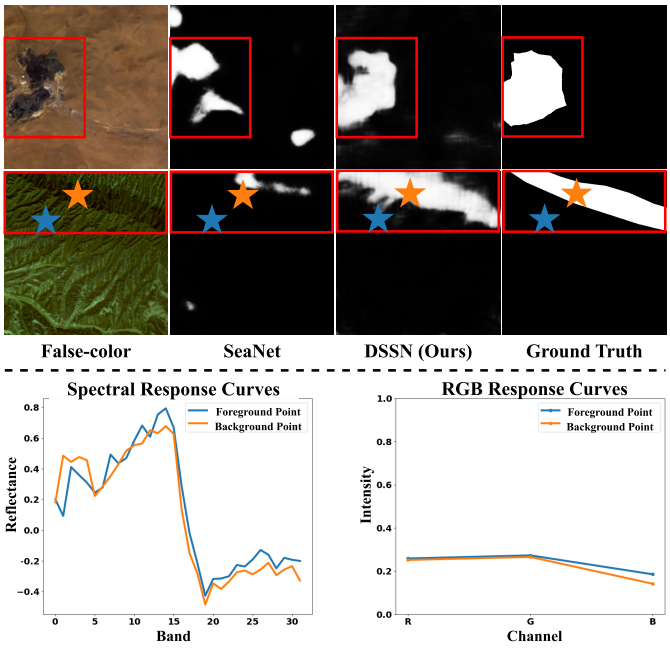


Fig. 1. Comparison with SeaNet, an RGB image-based method that struggles to detect salient objects when foreground (orange star) and background (blue star) colors are similar due to nearly uniform channel responses. In contrast, DSSN leverages rich spectral information for improved detection. The false-color image is synthesized by mapping three spectral bands (730, 580, and 466 nm) to the red, green, and blue channels, respectively.

3) *Multisalient objects:* The diversity and uneven spatial distribution of land cover types result in numerous images containing multiple salient objects, increasing detection complexity. These challenges provide a strong basis for developing and evaluating advanced HRSI-SOD methods.

The proposed methodology is motivated by the fundamental principle of "center-surround spectral difference evaluation," which evaluates the spectral similarity between central pixels and their surrounding neighbors to identify salient objects. Traditional methods [12], [13] predominantly rely on this principle. For instance, spectral saliency [13] constructs a pyramid structure by downsampling the input HSI and calculates spectral angle distances between pyramid layers to generate similarity maps. The final saliency map is produced by directly summing these individual maps. While this approach effectively implements "center-surround spectral difference evaluation," it suffers from three critical limitations.

1) *Limited multiscale adaptability:* Deep downsampling results in the loss of small targets, and the direct summation of multiscale similarity maps often causes larger targets to dominate, obscuring smaller ones.
2) *Inadequate foreground–background discrimination:* Relying solely on spectral information is insufficient for distinguishing foreground objects from background regions when their spectral characteristics are similar.
3) *Suboptimal handling of multiple objects:* Direct summation of similarity maps causes interference among multiple salient objects, leading to merged detections or missed small, scattered targets due to a lack of robust multiobject separation mechanisms.

To overcome these limitations, we propose the deep spectral saliency network (DSSN), an innovative baseline model for HRSI-SOD, which consists of three parts: spatial–spectral joint feature extractor (SJFE), cross-level saliency assessment block (CSAB), and high-resolution fusion module (HRFM). Each component is specifically designed to tackle one or more core limitations of traditional methods, enabling DSSN to robustly address large-scale variation, diverse foreground–background relations, and multisalient object challenges in HRSI-SOD.

The SJFE enhances foreground–background discriminability through parallel spatial and spectral branches. The spatial branch captures fine-grained textural details, while the spectral branch employs a spectral attention module to fully exploit spectral information. A hierarchical fusion mechanism adaptively integrates spatial and spectral features via cross-branch modulation, strengthening the representation of spatial–spectral features and improving foreground–background differentiation.

As the core structure of DSSN, the CSAB effectively implements "center-surround spectral difference evaluation." It first generates initial similarity maps by measuring cross-layer feature correlations. Then, it applies pixel-wise attention mechanisms to evaluate the contributions of multiscale similarity maps at each spatial location, enabling adaptive fusion through weighted aggregation. This fusion strategy suppresses erroneous responses in cluttered regions and emphasizes salient regions across scales, inherently addressing challenges related to large-scale variations and multiple salient objects.

The HRFM integrates multiscale intermediate saliency maps using a bottom–up fusion strategy, progressively combining high-resolution saliency maps with lower resolution ones. This strategy leverages the strengths of each scale: high-resolution maps capture small, scattered objects, while low-resolution maps capture larger ones. Additionally, learned spatial upsampling is employed to recover fine-grained details, ensuring accurate localization of small objects. This design ensures that DSSN can robustly detect multiple salient objects of varying sizes, even when they are closely spaced.

Experiments conducted on the HRSSD dataset underscore the critical importance of introducing a specialized benchmark for this domain. The results also highlight the high accuracy and computational efficiency of our DSSN. Further evaluations on additional datasets, including HSOD-BIT [2] and HS-SOD [9], confirm the strong generalizability of our approach. In a nutshell, our contributions can be summarized as follows.

1) We establish the first benchmark dataset for HRSI-SOD, introducing challenges such as large-scale variation, diverse foreground–background relations, and multisalient objects.
2) We propose a pioneering baseline model that specifically designed for hyperspectral remote sensing images SOD.
3) We introduce two innovative modules—CSAB and HRFM—that achieves accurate detection performance through pixel-wise attention mechanism and bottom-up multiscale fusion strategy, respectively.

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TABLE I
COMPARISON BETWEEN OUR HRSSD DATASET AND THE MAIN SOD DATASETS. STATISTICAL METRICS INCLUDE THE NUMBER OF IMAGES (#IMAGES), THE NUMBER OF SALIENT OBJECTS PER IMAGE (#OBJECTS), AND IMAGE RESOLUTION

Dataset	Data	Scene	Year	#Images	#Objects	Resolution
ECSSD [14]	Optical	Natural	2013	1000	~ 1	$\sim 400 \times 300$
DUT-OMRON [15]			2013	5168	~ 5	$\sim 400 \times 400$
PASCAL-S [16]			2014	850	~ 5	Variable
HKU-IS [17]			2015	4447	Multiple	$\sim 400 \times 300$
DUTS-TE [18]			2017	15572	Multiple	$\sim 400 \times 300$
ORSSD [19]	Optical	Remote Sensing	2019	800	Multiple	Variable
EORSSD [20]			2020	2000	Multiple	Variable
ORSI-4199 [21]			2022	4199	Multiple	Variable
RSISOD [22]			2023	5054	Multiple	Variable
RSSOD [23]			2023	6000	Multiple	Variable
HS-SOD [9]	Hyperspectral	Natural	2018	60	1	768×1024
HSOD-BIT [2]			2024	319	Multiple	1240×1680
HRSSD (Ours)	Hyperspectral	Remote Sensing	2025	704	Multiple	512×512

II. RELATED WORK

A. SOD Datasets

Significant progress in SOD has been driven by benchmark datasets with pixel-level annotations. For optical natural scenes, five predominant datasets are widely adopted: ECSSD [14], DUT-OMRON [15], PASCAL-S [16], HKU-IS [17], and DUTS-TE [18]. The development of SOD in optical remote sensing imagery remains nascent. Initial efforts include ORSSD [19] (800 images), followed by expanded collections like EORSSD [20] (2000 images) and ORSI-4199 [21] (4199 images). Recent datasets RSISOD [22] and RSSOD [23] specifically address challenges of scale variation, cluttered backgrounds, and seasonal diversity. However, these datasets fail to highlight the advantages of HSIs in detecting salient objects.

Imamoglu et al. [9] introduced the pioneering HS-SOD dataset, tailored for HSOD. The dataset does not include HSI-advantageous scenarios and presents few challenges. Qin et al. [2] proposed the HSOD-BIT dataset to address these issues. This dataset contains 319 natural scene images with foreground–background color similarity, overexposure, and uneven illumination challenges. Notably, no existing dataset targets SOD in hyperspectral remote sensing images. To fill this gap, we present the first benchmark specifically designed for HRSI-SOD, namely HRSSD. Key dataset characteristics are compared in Table I.

B. Hyperspectral SOD

Spectral saliency refers to the visual representation formed by aggregating pixels in HSIs with similar spectral characteristics that are visually distinguishable. This concept was first formalized by Le Moan et al. [24]. Early methods primarily built upon Itti's visual attention model [12]. For instance, Liang et al. [13] proposed a spectral pyramid approach that calculates spectral similarity through Euclidean or spectral angle distance metrics across multiple scales. Despite their computational efficiency, these methods are limited by hand-crafted features that struggle in complex scenes.

The advent of convolutional neural networks (CNNs) with advanced feature extraction capabilities has revolutionized this field. Imamoglu et al. [10] and Huang et al. [25] integrated CNNs with single/dual-branch architectures and clustering techniques (e.g. manifold ranking) to improve accuracy. To address clustering limitations, Liu et al. [1] proposed SMN, the first end-to-end trainable deep network. SMN incorporates specialized modules to extract high- and low-frequency components, along with mixed-frequency attention for enhanced feature utilization. In an effort to reduce annotation costs, Liu et al. [3] proposed the first weakly supervised pipeline for SOD in HSIs. However, these methods focus on natural scenes, with no existing solutions specifically addressing remote sensing scenarios.

C. Optical Remote Sensing Images SOD

Concurrent with HSOD advancements, optical remote sensing image SOD (ORSI-SOD) has undergone transformative evolution. Early ORSI-SOD methods relied on hand-crafted features that inherently limit detection robustness. The emergence of deep learning has fundamentally reshaped this landscape, with encoder–decoder architectures becoming the dominant paradigm. Landmark contributions include LV-Net [19]—the pioneering end-to-end fully convolutional framework with dual-stream pyramid modules for multiscale feature fusion. Subsequent innovations have focused on architectural enhancements. For instance, Li et al. [26] developed an adjacent context coordination module combining local saliency highlighting and global context integration from adjacent levels. Li et al. [27] introduced a correlation module with coarse-to-fine strategy, leveraging high-level semantic guidance for low-level feature localization. Li et al. [28] proposed dynamic semantic matching and edge self-alignment modules for cross-level feature processing. Gao et al. [29] pioneered Transformer integration through adaptive spatial tokenization encoders.

Despite these advances, existing ORSI-SOD methods remain incompatible with hyperspectral data. Bridging this domain gap, our proposed DSSN establishes the first dedicated

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baseline for HRSI-SOD. By preserving "center-surround spectral difference" principle, DSSN provides a foundational framework to advance this underexplored field.

III. HRSSD Benchmark Dataset

A. Dataset Construction

The HRSSD dataset was curated by carefully selecting 704 images from the WHU-OHS dataset [11], containing 5327 pixel-level annotated salient objects. The dataset is split into training and testing sets in a 5:2 ratio. Below, we provide a concise overview of the dataset construction process.

1) WHU-OHS Dataset: The WHU-OHS dataset is a comprehensive resource for land-cover classification using HSIs, comprising 7795 image patches captured by the Orbita hyperspectral satellite across over 40 locations in China. Covering an area exceeding 150000 km^2, each patch measures 512 × 512 pixels and includes 32 spectral bands spanning 466–940 nm. The dataset is labeled with 25 classes (0–24), where label 0 denotes ignored pixels, and labels 1–24 correspond to distinct land-cover categories. Our dataset retains the core attributes of WHU-OHS, and its authors have approved our reuse and redistribution.

2) Data Processing: In the original dataset, some images are segmented into training and testing subsets. We reassembled these segments to restore image integrity. Additionally, due to potential angular deviations in satellite-captured imagery, certain regions exhibit atypical radiance values. To mitigate this, we generated pixel-level masks for skewed regions, setting their pixel values to zero. This approach minimizes the influence of outliers, enhancing detection accuracy.

3) Data Filtering: We designate the category with the highest pixel count as the background, with all other categories classified as the foreground. Images were filtered based on spatial and spectral characteristics, followed by expert annotator voting.

Step 1: Spatial contrast filtering. Spatial contrast refers to the size difference between foreground and background. We prioritized images with significant spatial contrast and clear foreground–background boundary delineation.

Step 2: Spectral differentiation filtering: Effective HSOD requires substantial spectral contrast between foreground and background. We conducted spectral analysis to exclude images where spectral responses were too similar, as shown in Fig. 2(a), and retained those with distinct spectral differences, as in Fig. 2(b).

Step 3: Voting-Based Filtering: Annotators underwent 3 hours of training before participating in the filtering task. An image was included only if at least three out of five annotators agreed it met the criteria. Through this process, 806 images were initially selected, and after further expert review, 704 images were finalized to form the HRSSD dataset. The resulting dataset features distinct spectral differences between foreground and background, with foreground objects meeting salient object size requirements.

4) Salient Object Annotation: Background pixels are labeled as 0, foreground pixels as 1, and "ignore" pixels (e.g., boundaries with spectral mixing) as −1. Using this labeling strategy, we generated ground-truth images for the dataset.

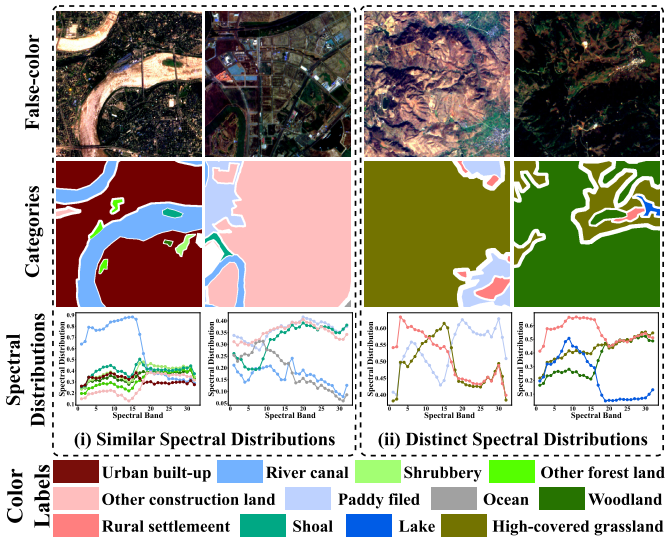


Fig. 2. Visualization of spectral statistics. To ensure significant spectral differences between the foreground and background, we excluded images with (a) similar spectral distributions and retained those with (b) distinct spectral distributions across different land-cover categories.

B. Statistics and Analysis

Based on the HRSSD dataset, we provide a comprehensive statistical analysis of salient object counts, object sizes, size bias, and center bias. These metrics enhance understanding of the dataset's characteristics and its challenges.

1) Number of Salient Objects Per Image: Among the 704 images in the dataset, 86 contain a single salient object, while the remaining images exhibit multiple salient objects, as depicted in Fig. 3(a). On average, each image contains 7.56 salient objects, underscoring the dataset's complexity and diversity in representing saliency.

2) Size of Salient Object: Salient objects in hyperspectral remote sensing imagery vary significantly in size, as depicted in Fig. 3(b). Fig. 3(a). illustrates the area ratio between the largest and smallest objects within an image, which can exceed 10 000 on a 512 × 512 image. Additionally, Fig. 3(d) shows the ratio of foreground to background pixels, revealing that approximately half of the images have a foreground-to-background pixel ratio below 0.1. The wide range of object sizes and the sparse presence of salient object pixels introduce significant challenges for SOD.

3) Size Bias and Center Bias: The width-to-height ratios of salient objects and their spatial distribution are analyzed in Fig. 3(e) and (f), respectively. These visualizations highlight substantial variation in object dimensions and their widespread locations within images. Notably, there is a tendency for salient objects to be concentrated near the image center, consistent with common SOD datasets [14], [15], [16], [17], [18]. This center bias reflects the natural tendency for important objects to appear centrally in images, further aligning the dataset with standard benchmarks.

C. Dataset Challenges

1) Large-Scale Variation: The HRSSD dataset exhibits significant variations in object sizes due to differences in land cover types and spatial scales, posing challenges in

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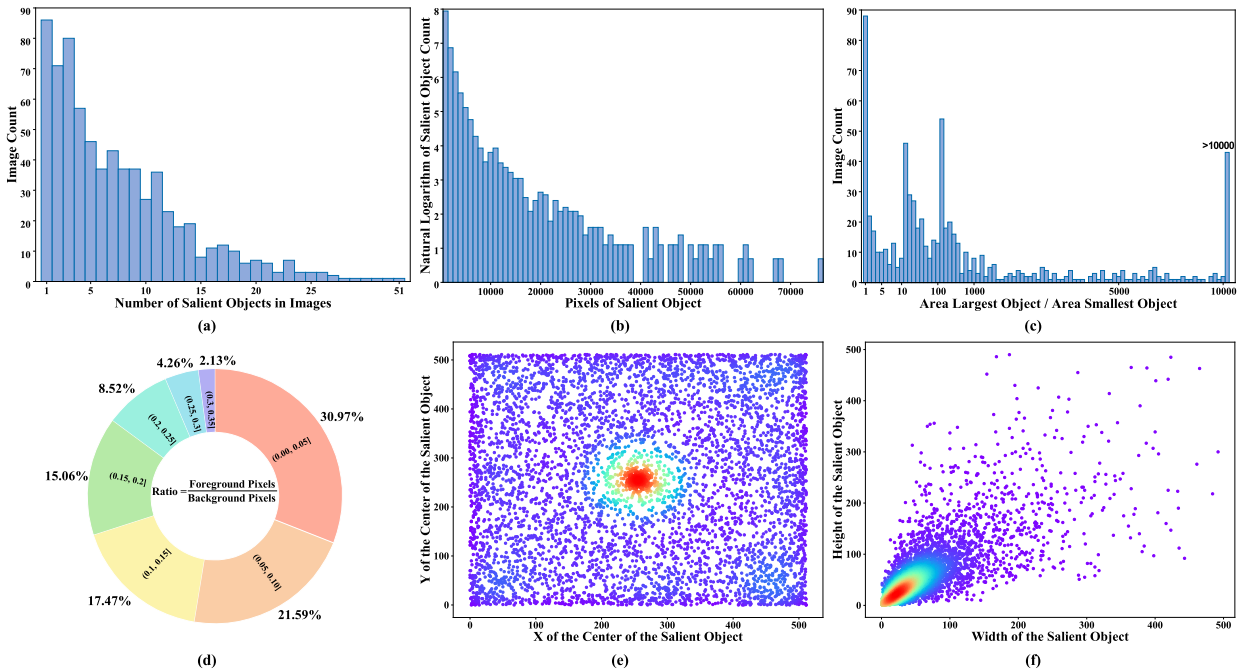


Fig. 3. Statistical overview of the HRSSD dataset. (a) Distribution of salient object count per image. (b) Size distribution of salient objects. (c) Area ratio between largest and smallest objects. (d) Image distribution by foreground-to-background pixel ratio. (e) Center bias in salient object placement. (f) Width and height biases in salient object sizes.

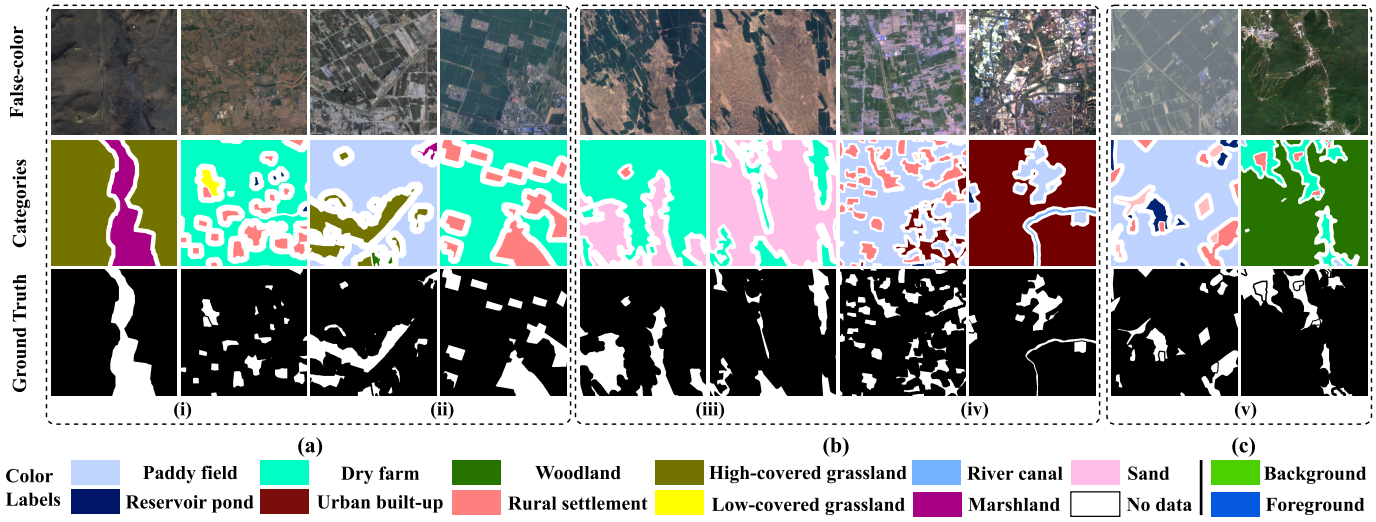


Fig. 4. Challenges in HRSSD. (i) and (ii) Illustrate (a) large-scale variations present both across different images and within a single image. The complete reversal of foreground–background separation in (iii) and (iv) underscores (b) diverse foreground–background relations. Many of the previous examples depict (c) multiple salient objects, with additional examples shown in (v).

two key aspects. For instance, the scale disparity between marshland and reservoir ponds [Fig. 4(a-i)] demonstrates cross-image scale variation, which occurs not only across different land cover types but also within the same type. This is further illustrated by high-coverage grassland and rural settlements [Fig. 4(a-ii)], which occupy varying spatial proportions within images. Statistical analyses in Fig. 3(b) and (c) confirm substantial scale variations both across and within images. These variations challenge SOD methods to robustly capture the overall contours of large-scale objects while maintaining precision in detailing small-scale ones.

2) Diverse Foreground–Background Relations: The spatial distribution and spectral characteristics of land cover, combined with its contextual variability, lead to dynamic foreground–background relationships across environments. For example, in Fig. 4(b-iii), river canals and surrounding marshlands switch roles as foreground and background in different scenes. Similarly, urban built-up areas and adjacent farmland exhibit reversed foreground–background relationships in Fig. 4(b-iv). This variability requires saliency detection algorithms to effectively distinguish foreground from background in diverse and complex scenarios, adding another layer of difficulty.

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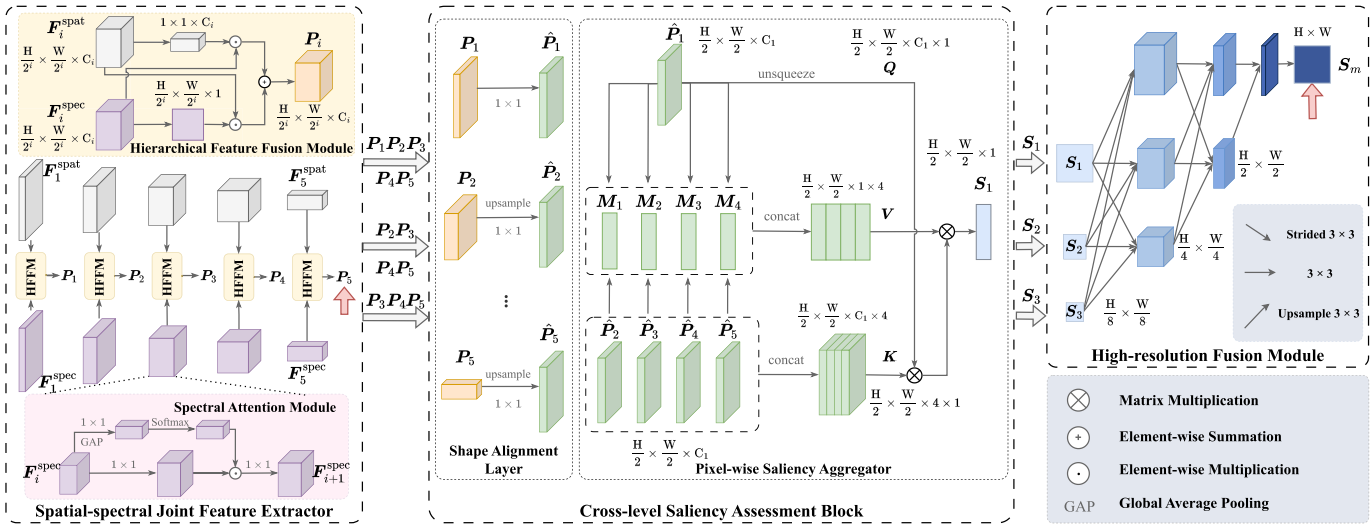


Fig. 5. Overview of the DSSN. The SJFE employs parallel branches to extract multilevel spatial–spectral features. The CSAB computes pixel-wise attention-based similarities across levels to generate multiscale similarity maps, which are then fused in the HRFM. The structure of the CSAB is illustrated with all features used as input. Red arrows indicate supervision signals.

3) Multisalient Objects: As shown in Fig. 4(b-v), over 90% of the images in HRSSD contain multiple salient objects, which may be closely positioned or exhibit diverse shapes. This complexity challenges the identification and segmentation of multiple salient objects within a single image.

IV. Deep Spectral Saliency Network

Given a HSI $\boldsymbol{I} \in \mathbb{R}^{H\times W\times C}$, the goal of SOD is to generate a saliency map $\boldsymbol{S}_{\mathrm{m}} \in \mathbb{R}^{H\times W\times 1}$ that highlights spectrally prominent objects or regions. This process can be expressed as

$$\boldsymbol{S}_{\mathrm{m}} = \boldsymbol{\Phi}(\boldsymbol{I}) \tag{1}$$

where $\boldsymbol{\Phi}(\cdot)$ denotes the mapping function that transforms the input HSI into a saliency map. Our proposed DSSN implements this mapping function. As depicted in Fig. 5, DSSN begins with the SJFE, which employs parallel branches to extract multilevel spatial and spectral features. Subsequently, the CSABs compute pixel-wise attention-based similarities across levels to produce multiscale similarity maps. Finally, the HRFM integrates these maps, enabling cross-resolution information exchange and generating the final saliency map.

A. Spatial–Spectral Joint Feature Extractor

To integrate spatial–spectral information inherent in HSIs and enhance feature representation, we propose a SJFE. As shown in Fig. 5, SJFE consists of two parallel branches (spatial branch and spectral branch) for multilevel feature extraction, followed by a Hierarchical feature fusion module to generate integrated spatial–spectral representations.

Spatial branch employs an off-the-shelf backbone network to extract hierarchical spatial features $\boldsymbol{F}^{\mathrm{spat}} = \{\boldsymbol{F}_i^{\mathrm{spat}}\}_{i=1}^{5}$. Each level i produces features $\boldsymbol{F}_i^{\mathrm{spat}} \in \mathbb{R}^{(H/2^i)\times(W/2^i)\times C_i}$, where H and W denote input dimensions, and C_i represents channel count at scale i.

Spectral branch integrates spectral attention modules to extract discriminative spectral features $\boldsymbol{F}^{\mathrm{spec}} = \{\boldsymbol{F}_i^{\mathrm{spec}}\}_{i=1}^{5}$ with dimension alignment to the spatial branch. It first generates channel-wise attention weights $\boldsymbol{v} \in \mathbb{R}^{C_i}$ through

$$\boldsymbol{v} = \sigma\big(\boldsymbol{f}_{\mathrm{GAP}}\big(\boldsymbol{f}_{\mathrm{C1}}(\boldsymbol{I})\big)\big). \tag{2}$$

The functions $\boldsymbol{f}_{\mathrm{C1}}(\cdot)$, $\boldsymbol{f}_{\mathrm{GAP}}(\cdot)$, and $\sigma(\cdot)$ represent a 1×1 convolution, global average pooling operation, and the Softmax activation, respectively. Features are then refined via

$$\boldsymbol{F}_1^{\mathrm{spec}} = \boldsymbol{f}_{\mathrm{C3}}\big(\boldsymbol{v}\odot\big(\boldsymbol{f}_{\mathrm{C2}}(\boldsymbol{I})\big)\big). \tag{3}$$

In this equation, $\boldsymbol{f}_{\mathrm{C2}}(\cdot)$ and $\boldsymbol{f}_{\mathrm{C3}}(\cdot)$ are 1×1 convolutions with distinct parameters, while $\odot$ indicates element-wise multiplication with broadcasting.

1) Hierarchical Feature Fusion: Cross-modal interaction is achieved through dual modulation at each pyramid level. Specifically, spatial features generate weight maps to recalibrate spectral features via

$$\hat{\boldsymbol{F}}_i^{\mathrm{spec}} = \boldsymbol{f}_{\mathrm{C4}}\big(\boldsymbol{F}_i^{\mathrm{spat}}\big)\odot\boldsymbol{F}_i^{\mathrm{spec}} \tag{4}$$

where $\boldsymbol{f}_{\mathrm{C4}}(\cdot)$ denotes a 1×1 convolution for channel adjustment. Meanwhile, spectral features produce channel weights through enhance spatial features

$$\hat{\boldsymbol{F}}_i^{\mathrm{spat}} = \boldsymbol{f}_{\mathrm{GAP}}\big(\boldsymbol{F}_i^{\mathrm{spec}}\big)\odot\boldsymbol{F}_i^{\mathrm{spat}}. \tag{5}$$

Final fusion combines modulated features through element-wise summation

$$\boldsymbol{P}_i = \hat{\boldsymbol{F}}_i^{\mathrm{spec}} \oplus \hat{\boldsymbol{F}}_i^{\mathrm{spat}} \tag{6}$$

where $\oplus$ signifies element-wise summation. $\boldsymbol{P}_i \in \mathbb{R}^{(H/2^i)\times(W/2^i)\times C_i}$ is the ith layer of the spatial–spectral joint feature pyramid $\boldsymbol{P} = \{\boldsymbol{P}_i\}_{i=1}^{5}$. These representations enable center-surround contrast analysis for saliency detection through subsequent multilevel feature comparison.

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B. Cross-Level Saliency Assessment Block

To conduct "center-surround spectral difference evaluation" and suppress erroneous responses in cluttered regions and emphasizes salient regions across scales, we introduce the CSAB, the central component of our framework. As depicted in Fig. 5, CSAB comprises two modules: the shape alignment layer for dimension standardization and the pixel-wise saliency aggregator for similarity evaluation and error suppression via pixel-wise attention mechanisms.

The proposed DSSN integrates three CSABs with shared architecture but varying input configurations. Each CSAB processes multilevel features to generate intermediate saliency maps $\boldsymbol{S} = \{\boldsymbol{S}_i\}_{i=1}^{3}$ at resolutions $(H/2^i) \times (W/2^i) \times 1$, where finer maps capture local details and coarser maps enhance object coverage.

1) Shape Alignment Layer: This layer aligns spatial and channel dimensions of input features through interpolation and linear transformation

$$\hat{\boldsymbol{P}}_i = \boldsymbol{f}_{\text{up}}(\boldsymbol{W}_i(\boldsymbol{P}_i)), \quad i = 1, 2, 3, 4, 5 \tag{7}$$

where $\boldsymbol{W}_i$ denotes 1×1 convolution for channel adjustment and $\boldsymbol{f}_{\text{up}}(\cdot)$ upsamples features to a unified spatial resolution of $(H/2) \times (W/2)$ with preserved channel count C_1.

2) Pixel-Wise Saliency Aggregator: Following Liang et al. [13], center-surround analysis is conducted by computing Euclidean distances between the highest-level feature $\hat{\boldsymbol{P}}_1$ and lower level features $\hat{\boldsymbol{P}}_{1+j}$

$$\boldsymbol{M}_j = \left\| \hat{\boldsymbol{P}}_1 - \hat{\boldsymbol{P}}_{1+j} \right\|, \quad j = 1, 2, 3, 4. \tag{8}$$

The obtained similarity maps $\boldsymbol{M}_j$ shares resolutions at $(H/2) \times (W/2) \times 1$. To suppress erroneous responses, a pixel-level attention mechanism adaptively weights similarity maps. For each pixel x, the top-level feature $\hat{\boldsymbol{P}}_1$ serves as the *query* $\boldsymbol{Q}(x) \in \mathbb{R}^{1\times1\times C_1}$, while lower level features $\hat{\boldsymbol{P}}_{1+j}$ provide *keys* $\boldsymbol{K}_j(x) \in \mathbb{R}^{1\times1\times C_1}$. Importance weights $\boldsymbol{A}(x) \in \mathbb{R}^{1\times1\times4}$ are computed as

$$\boldsymbol{A}(x) = \frac{\exp\left(\boldsymbol{Q}(x)^\top \boldsymbol{K}_j(x)\right)}{\sum_{\forall j} \exp\left(\boldsymbol{Q}(x)^\top \boldsymbol{K}_j(x)\right)}. \tag{9}$$

Such weights, serving as value, are applied to aggregate similarity maps into the final saliency map

$$\boldsymbol{S}_1(x) = \sum_{j=1}^{4} \boldsymbol{A}(x) \otimes \boldsymbol{V}_j(x) \tag{10}$$

where $\otimes$ denotes matrix multiplication. The implementation details are provided in Algorithm 1.

C. High-Resolution Fusion Module

As emphasized in [30] and [31], the fusion of multiscale features is crucial for detecting objects of varying sizes. To this end, we propose the HRFM, which employs a bottom–up fusion strategy to progressively aggregate information from high-resolution ($\boldsymbol{S}_1$) to low-resolution ($\boldsymbol{S}_3$) saliency maps, generating a final prediction with refined details. As shown in Fig. 5, HRFM consists of three cascaded fusion stages with decreasing channel counts.

Algorithm 1 Pytorch-Like Pseudo Code of CSAB

```
# feature_pyramid: input tensor
# q_index: a layer index
# k_indexes: list of layer indexes of the pyramid
# q_proj: a linear projection layer
# k_proj, v_proj: lists of linear projection layer
# dim: hidden_dim in CSAB

B, _, H, W = feature_pyramid[q_index].shape
Q = q_proj(feature_pyramid[q_index])
Ks, Vs~= [], []
for i in range(0, len(k_indexes)):
K = interpolate(feature_pyramid[k_indexes[i]],(H,W))
K = k_proj[i](rearrange(K,'b c h w -> b (h w) c'))
Ks.append(K)

V = interpolate(feature_pyramid[k_indexes[i]],(H,W))
V = v_proj[i](rearrange(V,'b c h w -> b (h w) c'))
Vs.append(pairwise_distance(Q,V,p=2).unsqueeze(-1))

Q = Q.view(B, -1, 1, dim)
K = cat(Ks, dim=-1).view(B, -1, len(k_indexes), dim)
V = cat(Vs, dim=-1).unsqueeze(-1)

attention_scores = matmul(Q, K.transpose(-1, -2))
attention_probs = softmax(attention_scores, dim=-1)
SMAP = matmul(attention_probs, V)
return rearrange(SMAP,'b (h w) 1~c -> b c h w', h=H)
```

Specifically, the first layer accepts three intermediate saliency maps $\{\boldsymbol{S}_l\}_{l=1}^{3}$ and generates larger sized features $\{\boldsymbol{F}_r^{\text{o}}\}_{r=1}^{3}$

$$\boldsymbol{F}_r^{\text{o}} = \sum_{l=1}^{3} \boldsymbol{f}_{lr}(\boldsymbol{S}_l) \tag{11}$$

where $\boldsymbol{f}_{lr}(\cdot)$ denotes resolution-specific transformations, dependent on the input index l and the output index r. When $l \geq r$, $\boldsymbol{f}_{lr}(\cdot)$ is implemented by bilinear interpolation followed by a convolution layer; when $l = r - 1$, $\boldsymbol{f}_{lr}(\cdot)$ consists of a single convolution layer; when $l = r - 2$, $\boldsymbol{f}_{lr}(\cdot)$ includes a convolution with a stride of 2. All convolution operations utilize 3×3 kernels, which extracts the spatial features more effectively, compared to 1×1 kernels.

Subsequent stages follow similar fusion patterns with reduced channel dimensions to balance computational efficiency, inversely mirroring the feature pyramid structure in the SJFE. This design ensures effective detection of multiple salient objects of varying sizes, even when they are closely spaced.

D. Objective Function

To optimize feature learning and saliency prediction, we employ dual supervision using standard loss functions applied to the final feature map $\boldsymbol{P}_5$ from the SJFE, and the ultimate saliency map $\boldsymbol{S}_{\text{m}}$. The training objective $\mathcal{L}$ combines two widely used loss functions $\mathcal{L}_{\text{BCE}}$ and $\mathcal{L}_{\text{IoU}}$

$$\begin{aligned}\mathcal{L} = {} & \mathcal{L}_{\text{BCE}}\left(\boldsymbol{P}_5', \boldsymbol{G}\right) + \mathcal{L}_{\text{IoU}}\left(\boldsymbol{P}_5', \boldsymbol{G}\right) \\ & + \mathcal{L}_{\text{BCE}}(\boldsymbol{S}_{\text{m}}, \boldsymbol{G}) + \mathcal{L}_{\text{IoU}}(\boldsymbol{S}_{\text{m}}, \boldsymbol{G})\end{aligned} \tag{12}$$

where $\boldsymbol{P}_5'$ is the final feature map $\boldsymbol{P}_5$ after channel alignment with the ground-truth saliency map $\boldsymbol{G}$. Binary cross-entropy

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(BCE) loss $\mathcal{L}_{\mathrm{BCE}}$ enforces pixel-wise accuracy

$$\mathcal{L}_{\mathrm{BCE}}(\boldsymbol{X}, \boldsymbol{G}) = -\sum \left[\boldsymbol{G}\log(\boldsymbol{X}) + (1-\boldsymbol{G})\log(1-\boldsymbol{X})\right]. \tag{13}$$

Meanwhile, intersection over union (IoU) loss $\mathcal{L}_{\mathrm{IoU}}$ enhances shape consistency

$$\mathcal{L}_{\mathrm{IoU}}(\boldsymbol{X}, \boldsymbol{G}) = 1 - \frac{\sum_{r=1}^{H}\sum_{c=1}^{W}\boldsymbol{X}(r,c)\boldsymbol{G}(r,c)}{\sum_{r=1}^{H}\sum_{c=1}^{W}[\boldsymbol{X}(r,c)+\boldsymbol{G}(r,c)-\boldsymbol{X}(r,c)\boldsymbol{G}(r,c)]} \tag{14}$$

where $\boldsymbol{X} \in \{\boldsymbol{P}_5', \boldsymbol{S}_{\mathrm{m}}\}$ denotes the network predictions, and H and W represent the height and width, respectively.

V. Experiments

We evaluate the performance of multiple methods using our HRSSD dataset and assess the efficacy of our DSSN on hyperspectral natural scene SOD datasets, HSOD-BIT [2] and HS-SOD [9].

A. Experimental Settings

1) Implementation Details: The proposed model is implemented on a single NVIDIA RTX 3090 GPU with an Intel Xeon Gold 5218R CPU. In the SJFE, MobileNetV2 [44] serves as the backbone for the spatial branch, while the spectral branch integrates our proposed spectral attention module as a replacement for MobileNetV2's inverted residual blocks. HSIs are uniformly resampled to a spatial resolution of 256×256 pixels. To prevent overfitting, data augmentation techniques including random horizontal flipping and cropping are applied. Following the protocol in [11], input values are scaled by a factor of $1/10\,000$ to preserve original radiometric characteristics instead of standard normalization. The model is optimized using the Nadam optimizer with an initial learning rate of 3×10^{-3}, which is dynamically adjusted via a cosine annealing schedule. Training is performed for 100 epochs with a batch size of 16 samples.

2) Competing Methods: As the first to propose a SOD method specifically tailored for hyperspectral remote sensing images, we have adapted recent HSI classification approaches to facilitate a comprehensive comparison. These include CSCN [39], SAHRNet [40], MambaHSI [41], MambaLG [42], and DSTC [43]. To further enrich our analysis, we incorporate three methods designed explicitly for hyperspectral natural scene images: SED [13], SG [13], SUDF [10], and the most recent open-source state-of-the-art approach, SMN [1].

To broaden the scope of our evaluation, we also include seven methods originally developed for optical remote sensing images: MJRBM [21], FSMINet [32], CorrNet [27], ACCoNet [26], SeaNet [28], and MEANet [33]. Additionally, we assess several methods designed for optical natural scene images, such as CTDNet [34], TRACER [35], BBRF [36], MENet [37], and ADMNet [38]. Methods designed for RGB images accept false-color images as input. All competing methods are implemented using their default settings.

3) Evaluation Metrics: Quantitative assessment employs four standardized metrics.

1) Mean absolute error (MAE) quantifies pixel-wise prediction accuracy through average absolute differences between saliency maps and ground truth.
2) Maximum F-measure (F_β) balances precision–recall trade-off via harmonic mean calculation.
3) Enhanced-alignment measure (E_ξ) evaluates both local pixel matching and global distribution consistency.
4) Structure measure (S_α) emphasizes spatial coherence by assessing object-level structural similarity.

B. Experiment on HRSSD

1) Quantitative Results: As demonstrated in Table II, our DSSN achieves state-of-the-art performance across multiple evaluation metrics on the HRSSD dataset. Specifically, DSSN outperforms competing methodologies in terms of F_β, E_ξ, and S_α. Although MENet achieves a marginally lower MAE score, its overall performance is limited by suboptimal scores in other metrics, particularly F_β. The inferior performance of methods designed for optical images underscores the critical importance of incorporating spectral information.

Methods tailored for hyperspectral natural scene images, such as SUDF and SMN, also exhibit limited effectiveness. While these methods leverage spectral information to some extent, they fail to capture the essence of "center-surround spectral difference evaluation," which is pivotal for addressing challenges such as large-scale variation, diverse foreground–background relations, and multisalient objects.

Despite significant progress in classification tasks, current HSI classification methods show limited adaptability to SOD. This limitation underscores the need for specialized algorithms and dedicated datasets specifically designed for HRSI-SOD.

2) Efficiency Analysis: In addition to superior detection performance, DSSN achieves an optimal balance between computational efficiency and accuracy. As shown in Table II, DSSN requires 4.95 G FLOPs and 5.15 million parameters, making it significantly more efficient than many competing methods. For example, ACCoNet and CSCN incur substantially higher computational costs (184.50 G and 286.50 G FLOPs, respectively), despite their relatively high MAE scores. Similarly, SUDF, while having the smallest parameter count (0.10 M), suffers from poor detection performance due to its reliance on minimal convolutional layers for feature extraction. DSSN's efficiency is further demonstrated by its inference speed of 43.12 FPS, surpassing most deep learning-based methods. Notably, while MambaLG achieves the highest inference speed (206.62 FPS), its detection performance is suboptimal, with an MAE of 0.075 and an F_β score of 0.491.

3) Qualitative Results: As illustrated in Fig. 6, DSSN demonstrates significant advantages in handling complex challenges inherent in the HRSSD dataset.

1) *Large-scale variation:* DSSN accurately detects both large-scale objects (e.g., rivers) and small-scale objects (e.g., rural settlements) with high precision. In contrast, methods like SMN and FSMINet struggle to comprehensively capture objects of varying scales.

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TABLE II
QUANTITATIVE RESULTS ON OUR HRSSD DATASET AND EFFICIENCY ANALYSIS

Method	Year	MAE ↓	F_β ↑	E_ξ ↑	S_α ↑	FLOPs (G)	#Params (M)	Speed (FPS)
*Methods for **Optical Remote Sensing Images***								
MJRBM [21]	2022	0.078	0.497	0.764	0.643	95.81	43.54	40.09
FSMINet [32]	2022	0.073	0.387	0.664	0.578	11.83	3.56	40.16
CorrNet [27]	2022	0.067	0.422	0.676	0.573	21.31	4.07	88.07
ACCoNet [26]	2023	0.064	0.487	0.722	0.629	184.50	102.55	55.12
SeaNet [28]	2023	0.075	0.408	0.713	0.580	1.81	2.75	76.28
MEANet [33]	2024	0.101	0.471	0.723	0.622	11.34	3.27	31.85
*Methods for **Optical Natural Scene Images***								
CTDNet [34]	2021	0.069	0.442	0.719	0.605	6.15	11.83	140.97
TRACER [35]	2022	0.066	0.466	0.735	0.621	2.17	3.90	13.37
BBRF [36]	2023	0.066	0.469	0.696	0.608	46.41	74.01	27.25
MENet [37]	2023	**0.063**	0.477	0.729	0.616	94.66	27.83	15.37
ADMNet [38]	2024	0.077	0.337	0.635	0.539	**0.87**	0.84	20.87
*Methods for **Hyperspectral Natural Scene Images***								
SED [13]	2013	0.108	0.281	0.621	0.518	-	-	2.24
SG [13]	2013	0.120	0.262	0.644	0.510	-	-	2.24
SUDF [10]	2019	0.221	0.148	0.479	0.449	82.90	**0.10**	0.51
SMN [1]	2024	0.084	0.331	0.652	0.546	14.58	7.27	35.91
*Methods for **Hyperspectral Remote Sensing Images***								
CSCN [39]	2024	0.069	0.526	0.767	0.644	286.50	7.38	23.78
SAHRNet [40]	2024	0.095	0.442	0.677	0.599	692.04	2.65	170.54
MambaHSI [41]	2024	0.080	0.488	0.749	0.636	18.95	0.556	206.12
MambaLG [42]	2025	0.075	0.491	0.724	0.630	94.65	1.55	**206.62**
DSTC [43]	2025	0.076	0.490	0.754	0.603	17.25	4.13	4.79
DSSN (Ours)	2025	0.069	**0.556**	**0.769**	**0.655**	4.95	5.15	43.12

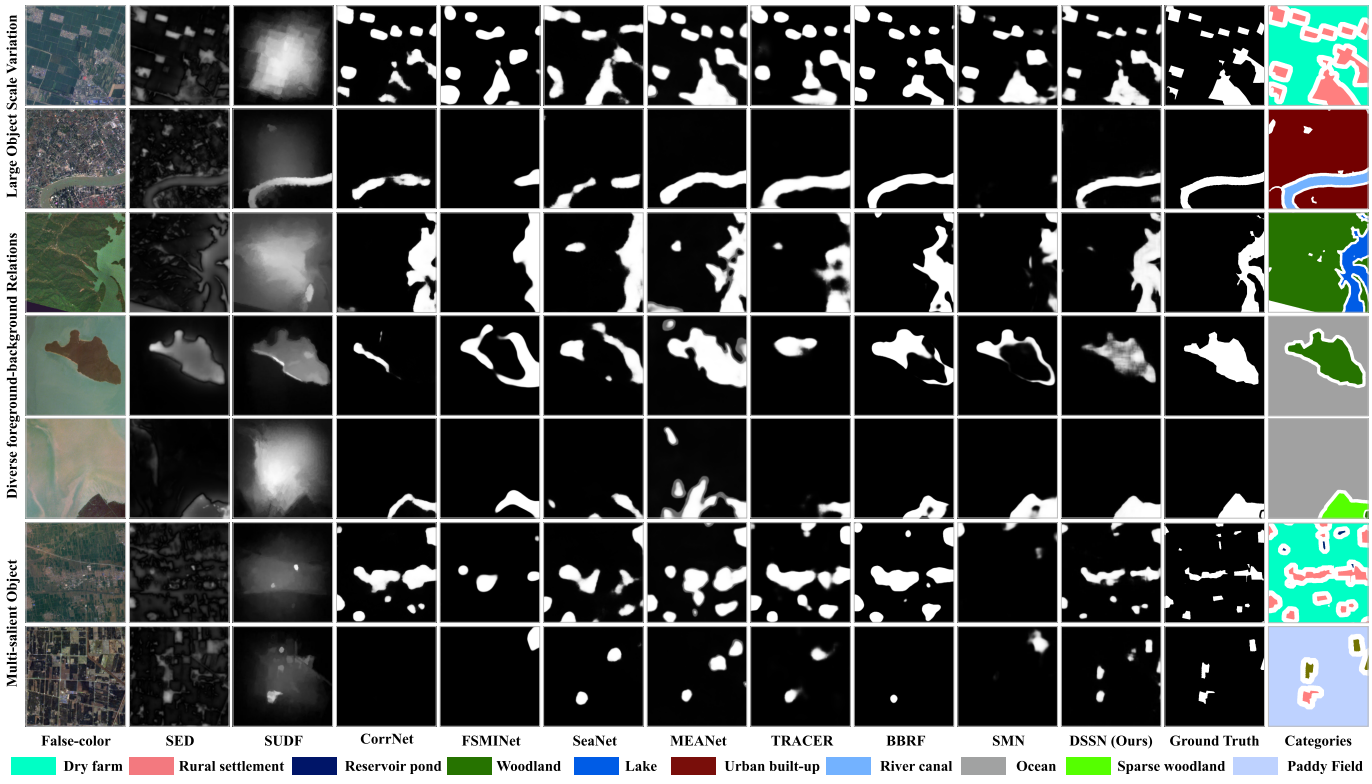


Fig. 6. Visualization results on the HRSSD dataset demonstrate the efficacy of our proposed DSSN. When confronted with the multiple challenges inherent in the dataset, our DSSN exhibits a substantial detection advantage over comparative methods.

2) *Diverse foreground–background relations:* DSSN effectively identifies salient objects regardless of their roles as foreground or background. For instance, it accurately detects woodland as the foreground against an

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TABLE III
QUANTITATIVE RESULTS ON HSOD-BIT AND HS-SOD DATASETS

Method	Year	HSOD-BIT				HS-SOD			
		MAE ↓	S_α ↑	AUC ↑	CC ↑	MAE ↓	S_α ↑	AUC ↑	CC ↑
SED [13]	2013	0.130	0.500	0.753	0.303	0.130	0.466	0.793	0.201
SG [13]	2013	0.182	0.543	0.791	0.370	0.171	0.521	0.808	0.268
SUDF [10]	2019	0.150	0.685	0.918	0.671	0.242	0.498	0.723	0.250
CTDNet [34]	2021	0.042	0.837	0.937	0.805	0.105	0.513	0.687	0.306
TRACER [35]	2022	0.039	0.862	**0.970**	0.846	0.157	0.595	0.868	0.465
BBRF [36]	2023	**0.033**	0.868	0.932	0.845	0.090	0.663	0.781	0.520
ADMNet [38]	2024	0.057	0.813	0.911	0.740	0.116	0.590	0.784	0.383
SMN [1]	2024	0.039	0.869	0.969	**0.849**	0.069	**0.767**	0.903	0.684
DSSN (Ours)	2025	0.035	**0.871**	**0.970**	0.840	**0.067**	0.755	**0.926**	**0.701**

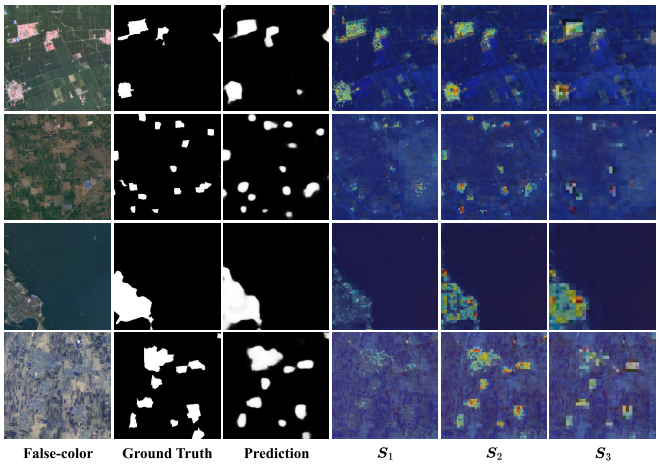


Fig. 7. Visualization of intermediate saliency maps.

ocean background and vice versa, whereas methods like TRACER and SeaNet perform inconsistently across such scenarios.

3) *Multisalient objects:* DSSN comprehensively detects all small salient objects in images with multiple targets, whereas other methods often miss some objects. Additionally, DSSN excels in detecting small scattered objects (e.g., rural settlements) and complex-shaped objects (e.g., woodland islands).

The superior performance of DSSN can be attributed to its innovative design components, specifically tailored to address the challenges in HRSI-SOD. The SJFE employs parallel spatial and spectral branches to enhance foreground–background discriminability by capturing complementary information from both domains. The CSAB adheres to the principle of "center-surround spectral difference evaluation" to generate initial similarity maps, while applying pixel-wise attention mechanisms to suppress erroneous responses in cluttered regions and emphasize salient regions across multiple scales. Additionally, the HRFM integrates a bottom–up fusion strategy with learned spatial upsampling to fully leverage the strengths of multiscale saliency maps, recovering fine-grained details and enabling the detection of multiple salient objects of varying sizes. These innovations collectively make DSSN highly effective for HRSI-SOD.

4) Visualization of Intermediate Saliency Maps: As shown in Fig. 7, the saliency maps generated by the CSABs reveal distinct characteristics at different resolutions. The high-resolution map $\boldsymbol{S}_1$ excels in capturing fine details of salient objects but is more prone to noise. In contrast, the lower resolution map $\boldsymbol{S}_3$ demonstrates superior accuracy in holistic SOD, albeit with less precision in fine details.

C. Experiment on HSOD-BIT and HS-SOD

1) Datasets: The HSOD-BIT dataset contains 319 HSIs of natural scenes, spanning a spectral range of 400 × 1000 nm with a spatial resolution of 1240 × 1680 pixels. It is divided into a training set of 255 images and a test set of 64 images. The dataset incorporates challenges such as similar foreground–background colors, overexposure, and uneven illumination to highlight the advantages of hyperspectral imaging under complex lighting conditions. The HS-SOD dataset comprises 60 HSIs with a spectral range of 380 × 780 nm sampled at 5 nm intervals and a spatial resolution of 768 × 1024 pixels. It is split into a training set of 48 images and a test set of 12 images. Both datasets provide corresponding RGB images and ground truth binary masks.

2) Implementation Details: For the HSOD-BIT dataset, we use a batch size of 48, while for HS-SOD, the batch size is set to 16. Training is conducted for 200 epochs across both datasets, with input images resized to 224 × 224 pixels.

3) Competing Methods: We compare our method with state-of-the-art approaches designed for hyperspectral natural scene images, including SED [13], SG [13], SUDF [10], and SMN [1]. Additionally, we evaluate it against methods developed for optical natural scene images, using false-color images generated from hyperspectral data as inputs. These include CTDNet [34], TRACER [35], BBRF [36], and ADMNet [38]. All competing methods are implemented using their default settings.

4) Evaluation Metrics: Performance is evaluated using four widely adopted metrics in HSOD tasks: MAE, structure measure (S_α), area under the curve (AUC), and correlation coefficient (CC). AUC reflects the model's ability to distinguish between salient and nonsalient regions, while CC measures the alignment between predicted saliency maps and ground truth.

5) Quantitative Results: As shown in Table III, DSSN consistently achieves low MAE values and high S_α, AUC, and CC scores across both datasets, demonstrating its effectiveness

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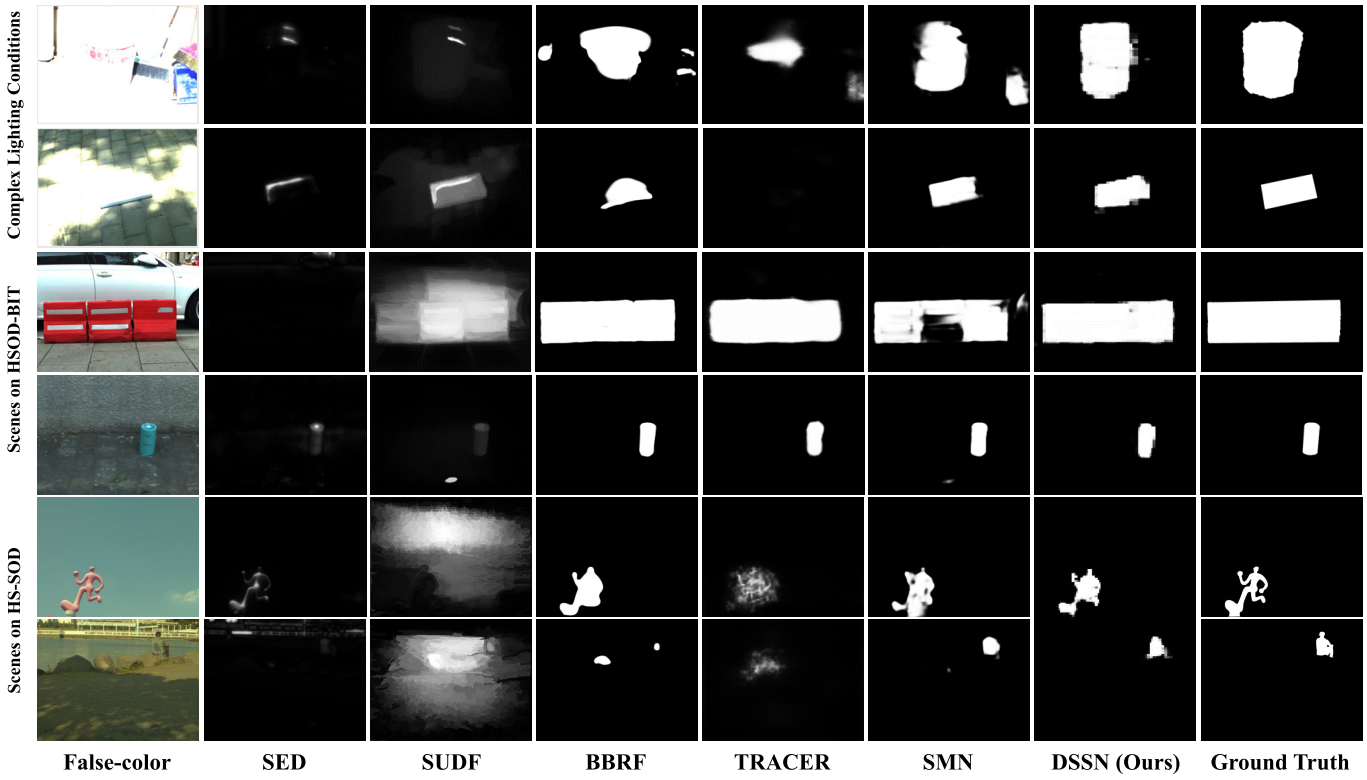


Fig. 8. Visualization results on HSOD-BIT and HS-SOD datasets. DSSN demonstrates superior detection performance in various scenarios.

and robustness. On the HSOD-BIT dataset, DSSN achieves an MAE of 0.035, outperforming all compared methods, with the closest competitor, BBRF, at 0.033. DSSN attains an S_α of 0.871, surpassing the best-performing baseline, SMN, which scores 0.869. It matches TRACER for the highest AUC at 0.970 and achieves a CC of 0.840, slightly below SMN's 0.849, indicating strong detection accuracy.

On the HS-SOD dataset, DSSN demonstrates robust performance with an MAE of 0.067, the lowest among all methods, outperforming SMN's second-best result of 0.069. For S_α, DSSN scores 0.755, second only to SMN's 0.767. DSSN achieves an AUC of 0.926, the second-highest after SMN's 0.903, and a CC of 0.701, showcasing competitive performance that exceeds most other methods.

6) Qualitative Results: As illustrated in Fig. 8, DSSN exhibits superior performance in SOD, particularly under complex lighting conditions and challenging scenes. Visualizations reveal that many methods struggle to accurately capture object boundaries, often resulting in blurred or incomplete detections. In contrast, DSSN provides precise boundary definitions and enhances object localization, closely resembling the ground truth.

On the HSOD-BIT dataset, unlike methods such as SED and SUDF, which frequently highlight irrelevant background regions or fail to produce high-contrast object areas, DSSN demonstrates minimal artifacts and achieves high accuracy. On the HS-SOD dataset, methods like SED and BBRF often struggle with finer structural details, resulting in fragmented or incomplete object contours. DSSN, however, effectively captures the entire object structure, producing outputs that closely align with the ground truth. Overall, DSSN demonstrates exceptional performance and robustness in addressing complex lighting conditions and diverse scenes, underscoring its superior generalization capability for hyperspectral natural scene images.

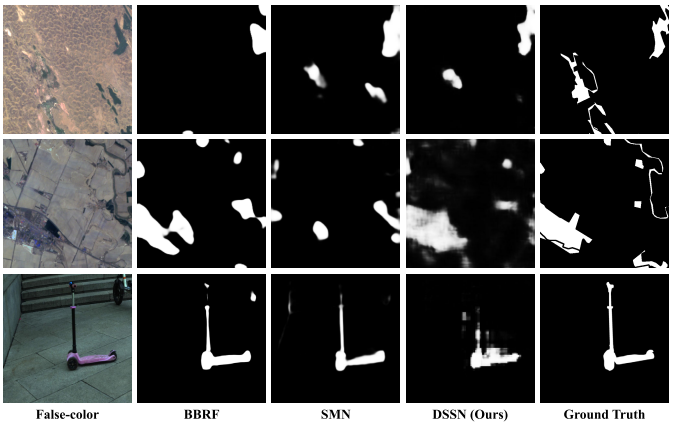


Fig. 9. Illustration of failure cases. DSSN and other comparative methods encounter challenges when target regions exhibit thin connecting parts.

D. Failure Cases

As shown in Fig. 9, the proposed DSSN, along with other comparative methods, demonstrates limitations in handling target regions with thin connecting parts. For example, in the first two remote sensing scenes, these methods struggle to accurately detect finer details, such as small land areas or narrow water bodies. In the final natural scene, DSSN incompletely

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TABLE IV
ABLATION STUDY OF MODALITIES IN FEATURE EXTRACTION

Spatial	Spectral	$F_\beta^{\max} \uparrow$	$S_\alpha \uparrow$
✓	✗	0.500	0.635
✗	✓	0.523	0.640
✓	✓	**0.556**	**0.655**

TABLE V
ABLATION STUDY OF CSAB AND HRFM

CSAB	HRFM	$F_\beta^{\max} \uparrow$	$S_\alpha \uparrow$
✗	✗	0.268	0.481
✓	✗	0.506	0.636
✗	✓	0.537	0.644
✓	✓	**0.556**	**0.655**

detects the scooter, capturing only its lower portion while failing to preserve the full structural details. Methods tailored for natural scenes, such as BBRF and SMN, achieve better performance.

E. Ablation Study

1) Effect of Input Modalities: Within the SJFE, we evaluated the performance of extracting spatial and spectral features individually and in combination. The results in Table IV demonstrate that simultaneous extraction of spatial and spectral features significantly outperforms single-modality approaches. Specifically, spatial feature extraction alone shows reductions of 0.023 and 0.005 in key metrics compared to spectral feature extraction alone. This highlights the critical contribution of spectrum to HRSI-SOD.

2) Effect of CSAB: The effectiveness of the CSAB is demonstrated in the second row of Table V. In this experiment, pyramid similarity maps were fused using direct summation instead of the proposed pixel-wise weighting scheme. By contrast, CSAB adaptively weights each similarity map, enabling more refined and context-aware fusion. This approach enhances the consolidation of salient features, leading to improved detection performance compared to the naive summation strategy.

3) Effect of HRFM: The third row of Table V evaluates HRFM. Here, HRFM was replaced with a series of stacked convolutional layers, maintaining equivalent layer count and feature dimensionality. This alternative configuration achieved F_β and S_α scores of 0.506 and 0.636, respectively, representing decreases of 0.050 and 0.019 compared to the original setup. These results underscore the importance of HRFM in preserving high-resolution features, facilitating inter-resolution information exchange, and effectively leveraging intermediate saliency maps for enhanced detection performance.

4) Ablation of Loss Functions: To assess the impact of different loss functions, we included the structural similarity index measure (SSIM) loss, which quantifies the structural similarity between predicted and ground truth maps. We performed an ablation study by removing each loss function individually to validate its effectiveness and examine the rationality of combining BCE Loss and IoU Loss. As shown in Table VI, the combination of BCE loss and IoU loss achieves superior performance, while the removal of any component leads to performance degradation. Notably, when using only SSIM Loss, the model failed to converge, and no valid results were obtained.

TABLE VI
ABLATION STUDY OF LOSS FUNCTIONS

BCE	IoU	SSIM	$F_\beta^{\max} \uparrow$	$S_\alpha \uparrow$
✓	✗	✗	0.503	0.623
✗	✓	✗	0.508	0.615
✗	✗	✓	-	-
✗	✓	✓	0.508	0.607
✓	✗	✓	0.448	0.602
✓	✓	✓	0.528	0.635
✓	✓	✗	**0.556**	**0.655**

VI. FUTURE WORK

Future research efforts could collect HSIs in a broader range of scenarios and explore advanced technologies such as data augmentation to increase the diversity and scale of datasets. Additionally, developing more sophisticated feature extraction methods capable of capturing fine-grained structural details will be critical for improving detection accuracy. Optimizing multiscale fusion strategies remains essential to preserve intricate details while ensuring precise identification of salient regions in complex scenes. Furthermore, exploring the practical applicability of these advancements in domains such as military defense, mineral mapping, and atmospheric monitoring could further enhance the model's robustness, adaptability, and generalization capabilities.

VII. CONCLUSION

In this work, we introduce the HRSSD, which includes 704 HSIs with 5327 pixel-level annotated salient objects. The dataset captures diverse challenges such as large-scale variations, complex foreground–background relationships, and multisalient object scenarios. Alongside HRSSD, we propose the DSSN, establishing a foundational baseline for HRSI-SOD. The three components of DSSN are designed to address aforementioned challenges, collectively enabling accurate and robust target detection. Experimental results demonstrate the necessity of HRSSD and highlight the effectiveness of DSSN in identifying salient objects across complex scenes.

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